Sachintha Ranasinghe

FULL-STACK ENGINEER | AI PRODUCT ENGINEERING

Dubai, United Arab Emirates | +971 55 434 7505 | **sachintha.ranasinghe.int@gmail.com**
LinkedIn: linkedin.com/in/sachintha-ranasinghe-620b572b4 | **GitHub:** github.com/SachinthaRanasinghe

PROFESSIONAL SUMMARY

Full-stack engineer building **AI-native web products** with React, Next.js, TypeScript, Node.js, and PostgreSQL. I ship systems with models inside them — multi-provider LLM routing, structured output, RAG, and browser automation — under test, with correctness enforced by architecture rather than by prompt. BSc (Hons) Software Engineering, First Class.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, SQL, Dart, Java, C#

Frontend: React 19, Next.js (App Router, Server Actions), Tailwind CSS, Vite, Radix UI, shadcn/ui

Backend: Node.js, Express, REST API design, Server-Sent Events, Django REST Framework, Flask

Databases: PostgreSQL, MySQL, SQLite, Firebase Firestore, schema design and migrations

AI / ML: LLM integration (Groq, Gemini, OpenAI-compatible APIs), RAG, tool calling, structured JSON output, prompt engineering, scikit-learn, LightGBM

Security & Auth: OAuth 2.0, refresh-token persistence, scoped API access, injection-safe rendering, input escaping

Testing & CI: Vitest, Playwright (end-to-end and automation), TDD, GitHub Actions, CI/CD

Graphics & Motion: Three.js, React Three Fiber, GLSL, Canvas API, Motion

Tools & Cloud: Git, GitHub, Vercel, serverless functions, Postman, Firebase, Netlify

SOFTWARE ENGINEERING PROJECTS

AI Job Application Agent

Next.js 16, React 19, TypeScript, SQLite, Playwright, Groq, Gemini

GitHub: **github.com/SachinthaRanasinghe/ai-job-application-agent**

- Built a local-first application that turns a job posting into a tailored resume and cover letter, then fills the employer's form and stops before submit — **57,000 lines of TypeScript across 16 modules and 126 test files**.
- Designed a **provenance guardrail**: every generated bullet carries the identifier of the profile item it derives from, so model-invented content is rejected structurally rather than filtered by prompt.
- Enforced a non-bypassable review gate in the schema and the test suite — content that was never approved cannot reach export.
- Routed extraction to Groq and tailoring to Gemini to **double the daily free-tier ceiling from 10 to 20 applications**, with no paid fallback and a loud failure instead.
- Reduced document preparation to a **1.9-minute median** from posting capture to two finished PDFs, measured across nine real applications.
- Automated employer forms with Playwright: ATS adapter detection, DOM introspection, field mapping, file attachment, and CAPTCHA hand-off.

SmartCareers.lk

React, TypeScript, Node.js, Express, PostgreSQL, Python

GitHub: **github.com/SachinthaRanasinghe/smartcareers-lk**

- Architected a platform serving **five distinct user roles** — job seekers, employers, students, universities, and administrators — with role-scoped workflows throughout.
- Implemented AI-powered job matching and recommendations, plus resume-building and skill-verification modules on Express and PostgreSQL services.
- Delivered a responsive interface across devices to widen access for students and job seekers.

Peak Blend Coffeehouse

JavaScript, Vercel serverless functions, Groq, SSE, Canvas

- Built a conversational barista that streams replies over **Server-Sent Events sentence by sentence**, cutting time to first audio.
- Implemented **LLM tool calling** so the assistant adds items to a live order, validated against a generated menu manifest.
- Wrote the chat endpoint against the OpenAI-compatible API shape, making the model provider a configuration change rather than a rewrite.

AI Daily Life Assistant

Flutter, Node.js, PostgreSQL, Groq API, Firebase

GitHub: **github.com/SachinthaRanasinghe/daily-life-ai-app**

- Engineered an AI-powered mobile app combining mood tracking, habits, finance management, and daily planning.
- Designed PostgreSQL-backed APIs and integrated Groq/Llama pipelines for personalized insights, finance suggestions, and adaptive schedules.
- Built the Flutter client with Riverpod state management and Hive on-device persistence.

Cafe Piranha Staff Management System

React, Vite, Firebase, Firestore, jsPDF

- Built and **deployed to production** a real-time staff attendance and payroll platform with GPS clock-ins, automated payroll, and live admin dashboards.
- Designed role-based workflows for session tracking, overtime approvals, and PDF/CSV reporting.

Through the Glass — 3D Portfolio

Next.js, React Three Fiber, GLSL, Zustand, Lenis

- Built a scroll-driven WebGL experience with custom GLSL shaders and a camera path driven from a Zustand store.
- Loaded the canvas client-only to eliminate an SSR hydration mismatch, and honored prefers-reduced-motion so scroll never locks.

Math Misconception Analyzer

Python, LightGBM, scikit-learn, Streamlit

- Built a machine learning application to classify student math misconceptions from written explanations.
- Trained a dual LightGBM pipeline reaching **MAP 0.712** on the Kaggle Student Math Misunderstanding metric.

EduMind AI

React, Django, Django REST Framework, PostgreSQL, Groq API

- Designed an adaptive learning platform with tutoring, automated grading, and student risk prediction across frontend, REST API, AI service, and analytics layers.

ADDITIONAL PROJECTS

AI Portfolio That Talks Back (Flask, Groq, Llama 3.3, RAG, Three.js)

Fluxora Cinematic Portfolio (React 19, TypeScript, Vite, Motion)

CabXpert Reservation System (Java, TDD, GitHub Actions)

Gallery Café Restaurant Management (PHP, MySQL, JavaScript) — **github.com/SachinthaRanasinghe/gallery-cafe**

Live Demo — **SachinthaRanasinghe.github.io/portfolio/**

EDUCATION

BSc (Hons) Software Engineering — First Class

Cardiff Metropolitan University (ICBT Campus, Sri Lanka) | 2024 - 2025 | Completed

Higher Diploma in Computing & Software Engineering — First Class

Cardiff Metropolitan University (ICBT Campus) | 2023 - 2024 | Completed

International Diploma in Information & Communication Technology — First Class

ICBT Campus | 2022 | Completed